

Artificial Intelligence & Machine Learning - Task 1

Task: Build & Evaluate a Linear Regression Model (House Price Predictor)

1. Objective

The objective of this project is to understand the complete Machine Learning workflow by building a Linear Regression model using the California Housing dataset. The project includes data loading, preprocessing, exploratory data analysis (EDA), model training, prediction, and evaluation.

The model predicts house prices based on various housing features such as income, house age, rooms, population, and location.

2. Dataset Description

The California Housing dataset is a popular dataset available in scikit-learn. It contains information collected from California districts. Each row represents one district and includes features related to houses and population.

Features used in the dataset:

MedInc	Median income	Population	Population in area
HouseAge	Average house age	AveOccup	Average occupancy
AveRooms	Average rooms	Latitude	Latitude location
AveBedrms	Average bedrooms	Longitude	Longitude location

Target Variable: HousePrice (Median house value)

Dataset Shape:
(20640, 9)

First 5 Rows:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	\
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	

	Longitude	HousePrice
0	-122.23	4.526
1	-122.22	3.585
2	-122.24	3.521
3	-122.25	3.413
4	-122.25	3.422

3. Data Preprocessing

Data preprocessing was performed before training the model. The dataset was checked for missing values and data types. No missing values were found in the dataset. Features and target variables were separated for model training.

The dataset was split into:

- 80% Training Data
- 20% Testing Data

The `train_test_split()` function from `scikit-learn` was used with `random_state=42` for reproducibility.

```
Missing Values:
MedInc      0
HouseAge    0
AveRooms    0
AveBedrms   0
Population  0
AveOccup    0
Latitude    0
Longitude   0
HousePrice  0
dtype: int64

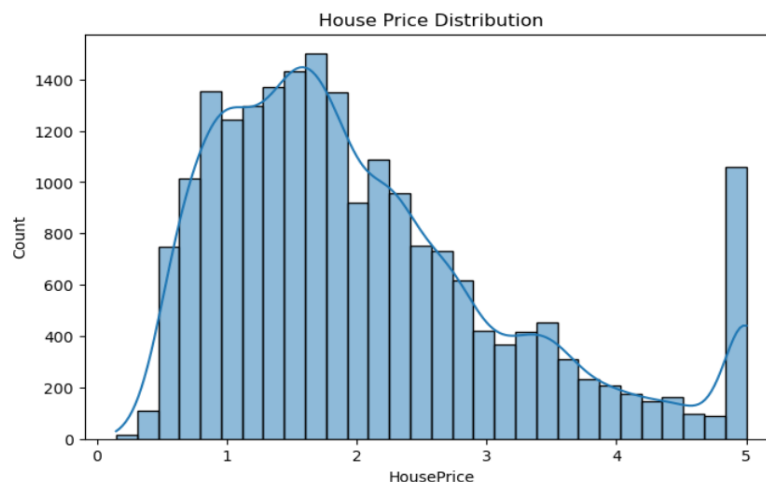
Training Data Shape: (16512, 8)
Testing Data Shape: (4128, 8)
```

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand patterns and relationships in the dataset. Different graphs and visualizations helped identify feature importance and trends.

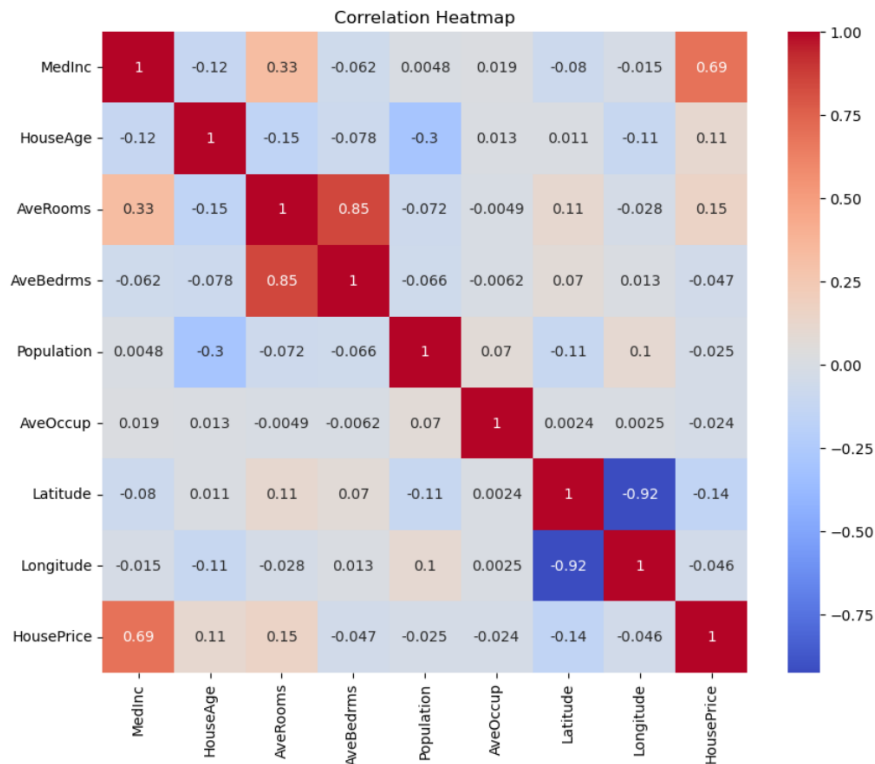
4.1 House Price Distribution

The histogram shows the distribution of house prices in the dataset. Most house prices are concentrated in the lower and medium price ranges.



4.2 Correlation Heatmap

The heatmap shows correlations between features. Median Income (MedInc) has a strong positive correlation with house prices, while some location-based features show negative correlation.



4.3 Median Income vs House Price

The scatter plot shows that higher median income generally leads to higher house prices. This indicates a positive relationship between income and house value.



5. Model Training

A Linear Regression model was trained using the training dataset. Linear Regression is a supervised Machine Learning algorithm used for predicting continuous values. The model learns the relationship between input features and target values using a best-fit line approach.

Steps Performed

- Import LinearRegression from sklearn.linear_model
- Train the model using .fit(X_train, y_train)
- Generate predictions using .predict(X_test)

```
# =====  
# TRAIN MODEL  
# =====  
  
model = LinearRegression()  
  
model.fit(X_train, y_train)  
  
print("\nModel Training Completed")
```

Model Training Completed

6. Model Evaluation

The trained model was evaluated using three important regression metrics:

- MAE (Mean Absolute Error) - measures the average prediction error.
- RMSE (Root Mean Squared Error) - measures prediction error while giving more importance to larger errors.
- R² Score - measures how well the model explains the variance in the data.

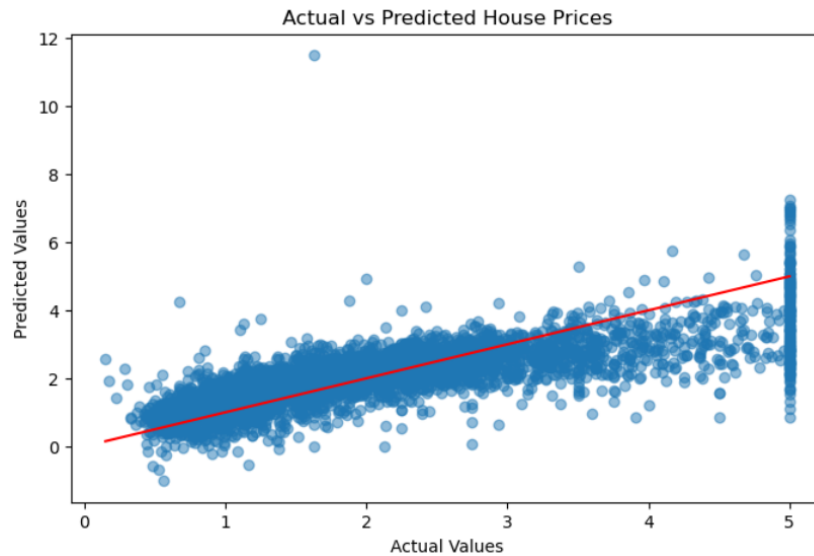
Results

Model Performance

MAE : 0.533
RMSE: 0.746
R2 Score: 0.576

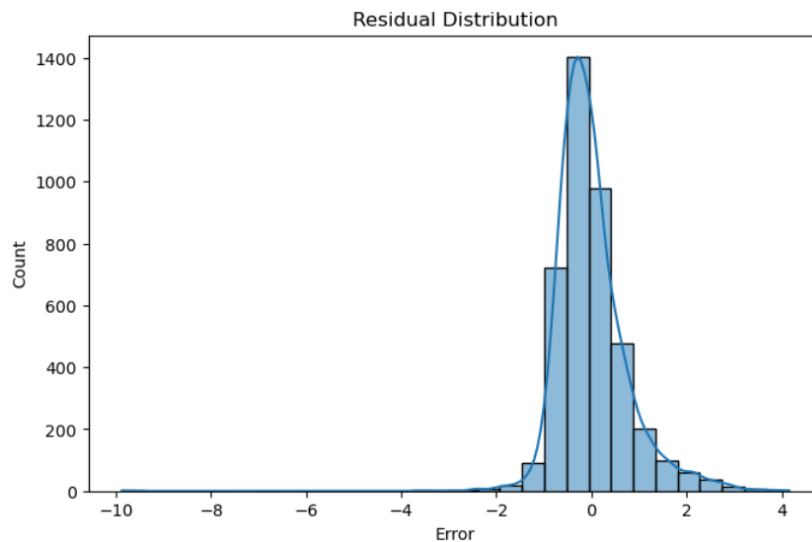
7. Actual vs Predicted Values

The Actual vs Predicted graph compares the real house prices with predicted house prices. Most points are close to the red diagonal line, indicating good prediction performance.



8. Residual Error Distribution

Residual analysis helps understand prediction errors. The residual graph shows that most errors are distributed around zero, which indicates reasonable model performance.



9. Conclusion

This project successfully implemented a Linear Regression model using the California Housing dataset. The model was able to predict house prices with reasonable accuracy.

Key Learnings

- Data preprocessing techniques
- Data visualization and exploratory analysis

- Model training with scikit-learn
- Prediction techniques using Linear Regression
- Model evaluation using MAE, RMSE, and R^2 Score

Linear Regression works well for basic prediction tasks and provides a strong foundation for advanced Machine Learning models.